DBMS LAB DAY-5

LAB ASSIGNMENT–5

Question 1. Create the following tables (naming all constraints is mandatory).

A. CLIENT_MASTER

Column Name	Data Type	Size	Constraint
Client_no	VARCHAR2	5	Primary key; must start with C
Name	VARCHAR2	20	NOT NULL, UNIQUE
Address1	VARCHAR2	30	—
State	VARCHAR2	30	—
City	VARCHAR2	15	Must be Delhi, Mumbai or Chennai

B. SALES_ORDER

Column Name	Data Type	Size	Constraint
S_order_no	VARCHAR2	10	Primary key; must start with O
S_order_date	DATE	—	—
Client_no	VARCHAR2	5	Foreign key to CLIENT_MASTER(Client_no)
Salesman_no	VARCHAR2	10	Must start with S
Product_no	VARCHAR2	10	Foreign key to PRODUCTS_MASTER(Product_no)

C. PRODUCTS_MASTER

Column Name	Data Type	Size	Constraint
Product_no	VARCHAR2	10	Primary key; must start with P
Description	VARCHAR2	20	NOT NULL, UNIQUE
Qty_on_hand	NUMBER	8	Must be greater than 10
Sell_price	NUMBER	8	2
Cost_price	NUMBER	8	2

Create PRODUCTS_MASTER first so SALES_ORDER can reference it; then create CLIENT_MASTER and SALES_ORDER.

```
CREATE TABLE PRODUCTS_MASTER (
  Product_no   VARCHAR2(10) CONSTRAINT PM_PK PRIMARY KEY
               CONSTRAINT PM_NO_CK CHECK (Product_no LIKE 'P%'),
  Description  VARCHAR2(20) CONSTRAINT PM_DESC_NN NOT NULL
               CONSTRAINT PM_DESC_UK UNIQUE,
  Qty_on_hand  NUMBER(8) CONSTRAINT PM_QTY_CK CHECK (Qty_on_hand > 10),
  Sell_price   NUMBER(8,2) CONSTRAINT PM_SELL_NN NOT NULL,
  Cost_price   NUMBER(8,2) CONSTRAINT PM_COST_NN NOT NULL
);

CREATE TABLE CLIENT_MASTER (
  Client_no VARCHAR2(5) CONSTRAINT CM_PK PRIMARY KEY
            CONSTRAINT CM_NO_CK CHECK (Client_no LIKE 'C%'),
  Name      VARCHAR2(20) CONSTRAINT CM_NAME_NN NOT NULL
            CONSTRAINT CM_NAME_UK UNIQUE,
  Address1  VARCHAR2(30),
  State     VARCHAR2(30),
  City      VARCHAR2(15) CONSTRAINT CM_CITY_CK
            CHECK (City IN ('Delhi','Mumbai','Chennai'))
);

CREATE TABLE SALES_ORDER (
  S_order_no   VARCHAR2(10) CONSTRAINT SO_PK PRIMARY KEY
               CONSTRAINT SO_NO_CK CHECK (S_order_no LIKE 'O%'),
  S_order_date DATE,
```

```
    Client_no    VARCHAR2(5) CONSTRAINT SO_CLIENT_FK
                 REFERENCES CLIENT_MASTER(Client_no),
    Salesman_no  VARCHAR2(10) CONSTRAINT SO_SALESMAN_CK
                 CHECK (Salesman_no LIKE 'S%'),
    Product_no   VARCHAR2(10) CONSTRAINT SO_PRODUCT_FK
                 REFERENCES PRODUCTS_MASTER(Product_no)
  );
```

Output: 3 tables created successfully.

Insert the supplied records

```
-- PRODUCTS_MASTER
INSERT INTO PRODUCTS_MASTER VALUES ('P01','1.44 Floppies',100,525,500);
INSERT INTO PRODUCTS_MASTER VALUES ('P02','Monitors',25,12000,11280);
INSERT INTO PRODUCTS_MASTER VALUES ('P03','Mouse',20,1050,1000);
INSERT INTO PRODUCTS_MASTER VALUES ('P04','1.22 floppies',100,525,500);
INSERT INTO PRODUCTS_MASTER VALUES ('P05','Keyboards',15,3150,3050);
INSERT INTO PRODUCTS_MASTER VALUES ('P06','Cd drive',14,5250,5100);

-- CLIENT_MASTER
INSERT INTO CLIENT_MASTER VALUES ('C01','Ivaan','Church Rd','Maharashtra','Mumbai');
INSERT INTO CLIENT_MASTER VALUES ('C02','Vandana','St.Mary Rd','Tamil Nadu','Chennai');
INSERT INTO CLIENT_MASTER VALUES ('C03','Pramada','Mall Rd','Maharashtra','Mumbai');
INSERT INTO CLIENT_MASTER VALUES ('C04','Basu','Church Rd','Maharashtra','Mumbai');
INSERT INTO CLIENT_MASTER VALUES ('C05','Ravi','Chandni',NULL,'Delhi');
INSERT INTO CLIENT_MASTER VALUES ('C06','Rukmini','Mall Rd','Maharashtra','Mumbai');

-- SALES_ORDER
INSERT INTO SALES_ORDER VALUES ('O19001',TO_DATE('12-jan-96','DD-MON-RR'),'C01','S01','P01');
INSERT INTO SALES_ORDER VALUES ('O19002',TO_DATE('25-jan-96','DD-MON-RR'),'C02','S02','P02');
INSERT INTO SALES_ORDER VALUES ('O19003',TO_DATE('18-feb-96','DD-MON-RR'),'C03','S03','P03');
INSERT INTO SALES_ORDER VALUES ('O19004',TO_DATE('03-apr-96','DD-MON-RR'),'C01','S01','P04');
INSERT INTO SALES_ORDER VALUES ('O19005',TO_DATE('20-may-96','DD-MON-RR'),'C04','S02','P05');
INSERT INTO SALES_ORDER VALUES ('O19006',TO_DATE('24-may-96','DD-MON-RR'),'C05','S04','P06');

COMMIT;
```

Output: 6 rows inserted into each table (18 rows total).

Initial table contents

Client_no	Name	Address1	State	City
C01	Ivaan	Church Rd	Maharashtra	Mumbai
C02	Vandana	St.Mary Rd	Tamil Nadu	Chennai
C03	Pramada	Mall Rd	Maharashtra	Mumbai
C04	Basu	Church Rd	Maharashtra	Mumbai
C05	Ravi	Chandni	NULL	Delhi
C06	Rukmini	Mall Rd	Maharashtra	Mumbai

S_order_no	S_order_date	Client_no	Salesman_no	Product_no
O19001	12-Jan-96	C01	S01	P01
O19002	25-Jan-96	C02	S02	P02
O19003	18-Feb-96	C03	S03	P03
O19004	03-Apr-96	C01	S01	P04
O19005	20-May-96	C04	S02	P05
O19006	24-May-96	C05	S04	P06

Product_no	Description	Qty_on_hand	Sell_price	Cost_price
P01	1.44 Floppies	100	525	500
P02	Monitors	25	12000	11280
P03	Mouse	20	1050	1000
P04	1.22 floppies	100	525	500
P05	Keyboards	15	3150	3050
P06	Cd drive	14	5250	5100

2. Add a NOT NULL constraint on Address1 in CLIENT_MASTER and display the table structure.

SQL Query:

```
ALTER TABLE CLIENT_MASTER MODIFY (Address1 CONSTRAINT CM_ADDRESS_NN NOT NULL);
DESC CLIENT_MASTER;
```

All existing Address1 values are non-NULL, so the constraint can be added.

Output:

```
Output: Table altered.\nCLIENT_NO  VARCHAR2(5)   NOT NULL\nNAME       VARCHAR2(20)  NOT NULL\nADDRESS1   VARCHAR2(30)  NOT
NULL\nSTATE      VARCHAR2(30)\nCITY       VARCHAR2(15)
```

3. Calculate profit (Sell_price - Cost_price) from PRODUCTS_MASTER. Name the column Profit.

SQL Query:

```
SELECT Product_no, Description, Sell_price - Cost_price AS Profit FROM PRODUCTS_MASTER ORDER BY Product_no;
```

Output:

PRODUCT_NO	DESCRIPTION	PROFIT
P01	1.44 Floppies	25
P02	Monitors	720
P03	Mouse	50
P04	1.22 floppies	25
P05	Keyboards	100
P06	Cd drive	150

4. Calculate and display total cost price (Qty_on_hand * Cost_price) of stock present. Name the column accordingly.

SQL Query:

```
SELECT SUM(Qty_on_hand * Cost_price) AS TOTAL_STOCK_COST FROM PRODUCTS_MASTER;
```

Output:

TOTAL_STOCK_COST
1272700

5. Display client details of all clients whose name starts with I.

SQL Query:

```
SELECT * FROM CLIENT_MASTER WHERE Name LIKE 'I%';
```

Output:

CLIENT_NO	NAME	ADDRESS1	STATE	CITY
C01	Ivaan	Church Rd	Maharashtra	Mumbai

6. Display client details of all clients whose name starts with R and ends with i.

SQL Query:

```
SELECT * FROM CLIENT_MASTER WHERE Name LIKE 'R%i';
```

Correction: the LIKE pattern R%i matches both Ravi and Rukmini because both start with R and end with i.

Output:

CLIENT_NO	NAME	ADDRESS1	STATE	CITY
C05	Ravi	Chandni	NULL	Delhi
C06	Rukmini	Mall Rd	Maharashtra	Mumbai

7. Display client details of all clients whose name contains a in the third and fifth position.

SQL Query:

```
SELECT * FROM CLIENT_MASTER WHERE Name LIKE '__a_a%';
```

Output:

CLIENT_NO	NAME	ADDRESS1	STATE	CITY
C03	Pramada	Mall Rd	Maharashtra	Mumbai

8. Display client details of all clients whose name contains aa.

SQL Query:

```
SELECT * FROM CLIENT_MASTER WHERE Name LIKE '%aa%';
```

Output:

CLIENT_NO	NAME	ADDRESS1	STATE	CITY
C01	Ivaan	Church Rd	Maharashtra	Mumbai

9. Display client details of all clients whose name contains exactly four characters.

SQL Query:

```
SELECT * FROM CLIENT_MASTER WHERE LENGTH(Name) = 4;
```

Output:

CLIENT_NO	NAME	ADDRESS1	STATE	CITY
C04	Basu	Church Rd	Maharashtra	Mumbai
C05	Ravi	Chandni	NULL	Delhi

10. Display client details of those clients who have not mentioned state.

SQL Query:

```
SELECT * FROM CLIENT_MASTER WHERE State IS NULL;
```

Output:

CLIENT_NO	NAME	ADDRESS1	STATE	CITY
C05	Ravi	Chandni	NULL	Delhi

11. Display order details placed after January 1996.

SQL Query:

```
SELECT * FROM SALES_ORDER
WHERE S_order_date >= DATE '1996-02-01'
ORDER BY S_order_date;
```

Output:

S_ORDER_NO	S_ORDER_DATE	CLIENT_NO	SALESMAN_NO	PRODUCT_NO
O19003	18-Feb-96	C03	S03	P03
O19004	03-Apr-96	C01	S01	P04
O19005	20-May-96	C04	S02	P05
O19006	24-May-96	C05	S04	P06

12. Change the order date of client C01 to 24/07/96, Product_no to P06 and Salesman_no to S04.

SQL Query:

```
UPDATE SALES_ORDER
SET S_order_date = TO_DATE('24/07/96','DD/MM/RR'),
    Product_no = 'P06',
    Salesman_no = 'S04'
WHERE Client_no = 'C01';
SELECT * FROM SALES_ORDER WHERE Client_no = 'C01';
```

Two rows are updated because two orders belong to client C01. Output: 2 rows updated.

Output:

S_ORDER_NO	S_ORDER_DATE	CLIENT_NO	SALESMAN_NO	PRODUCT_NO
O19001	24-Jul-96	C01	S04	P06
O19004	24-Jul-96	C01	S04	P06

13. Change the city of client C05 to Kolkata.

SQL Query:

```
UPDATE CLIENT_MASTER SET City = 'Kolkata' WHERE Client_no = 'C05';
SELECT Client_no, Name, City FROM CLIENT_MASTER WHERE Client_no = 'C05';
```

The original city CHECK constraint allows only Delhi, Mumbai and Chennai. To perform this requested update, first change the constraint to include Kolkata:

Output:

CLIENT_NO	NAME	CITY
C05	Ravi	Kolkata

14. Change the field size of Client_no to 15 in all tables where Client_no is present.

SQL Query:

```
-- Drop the foreign key temporarily, alter both columns, then recreate it.
ALTER TABLE SALES_ORDER DROP CONSTRAINT SO_CLIENT_FK;
ALTER TABLE CLIENT_MASTER MODIFY (Client_no VARCHAR2(15));
ALTER TABLE SALES_ORDER MODIFY (Client_no VARCHAR2(15));
ALTER TABLE SALES_ORDER ADD CONSTRAINT SO_CLIENT_FK
  FOREIGN KEY (Client_no) REFERENCES CLIENT_MASTER(Client_no);
```

Output:

```
Output: Table altered (four ALTER TABLE operations).
```

15. Remove the record for Client_no C02 from CLIENT_MASTER.

SQL Query:

```
-- The sales order O19002 refers to C02. Delete the child row first.
DELETE FROM SALES_ORDER WHERE Client_no = 'C02';
DELETE FROM CLIENT_MASTER WHERE Client_no = 'C02';
COMMIT;
SELECT * FROM CLIENT_MASTER ORDER BY Client_no;
```

Output: 1 sales-order row and 1 client row deleted. The child row must be removed first to satisfy the foreign-key constraint.

Output:

CLIENT_NO	NAME
C01	Ivaan
C03	Pramada
C04	Basu
C05	Ravi
C06	Rukmini

16. Remove records from PRODUCTS_MASTER for which Sell_price is between 1000 and 10,000.

SQL Query:

```
-- Delete sales orders that reference the products before deleting those products.
DELETE FROM SALES_ORDER WHERE Product_no IN ('P03','P05','P06');
DELETE FROM PRODUCTS_MASTER WHERE Sell_price BETWEEN 1000 AND 10000;
COMMIT;
SELECT Product_no, Description, Sell_price FROM PRODUCTS_MASTER ORDER BY Product_no;
```

Products P03, P05 and P06 are deleted. Their referencing sales orders are deleted first to avoid foreign-key violations. P02 remains because its sell price is 12,000.

Output:

PRODUCT_NO	DESCRIPTION	SELL_PRICE
P01	1.44 Floppies	525
P02	Monitors	12000
P04	1.22 floppies	525

17. Create a table of your own with a composite primary key.

SQL Query:

```
CREATE TABLE STUDENT_SUBJECT (
  Student_id   NUMBER(5) CONSTRAINT SS_STUDENT_NN NOT NULL,
  Subject_code VARCHAR2(10) CONSTRAINT SS_SUBJECT_NN NOT NULL,
  Marks        NUMBER(3),
  CONSTRAINT SS_PK PRIMARY KEY (Student_id, Subject_code)
);
INSERT INTO STUDENT_SUBJECT VALUES (101,'DBMS',85);
INSERT INTO STUDENT_SUBJECT VALUES (101,'JAVA',90);
INSERT INTO STUDENT_SUBJECT VALUES (102,'DBMS',78);
SELECT * FROM STUDENT_SUBJECT ORDER BY Student_id, Subject_code;
```

The combination (Student_id, Subject_code) is the composite primary key. Repeating either column alone is allowed; repeating the pair is not.

Output:

STUDENT_ID	SUBJECT_CODE	MARKS
101	DBMS	85
101	JAVA	90
102	DBMS	78

18. Create another table where the composite primary key from Question 17 acts as a foreign key.

SQL Query:

```
CREATE TABLE STUDENT_SUBJECT_RESULT (
  Result_id    NUMBER(5) CONSTRAINT SSR_RESULT_PK PRIMARY KEY,
  Student_id   NUMBER(5) NOT NULL,
  Subject_code VARCHAR2(10) NOT NULL,
  Grade        VARCHAR2(2),
  CONSTRAINT SSR_SS_FK FOREIGN KEY (Student_id, Subject_code)
    REFERENCES STUDENT_SUBJECT (Student_id, Subject_code)
);
INSERT INTO STUDENT_SUBJECT_RESULT VALUES (1,101,'DBMS','A');
INSERT INTO STUDENT_SUBJECT_RESULT VALUES (2,101,'JAVA','A+');
INSERT INTO STUDENT_SUBJECT_RESULT VALUES (3,102,'DBMS','B');
SELECT * FROM STUDENT_SUBJECT_RESULT ORDER BY Result_id;
```

The foreign key contains both columns of the composite primary key in STUDENT_SUBJECT.

Output:

RESULT_ID	STUDENT_ID	SUBJECT_CODE	GRADE
1	101	DBMS	A
2	101	JAVA	A+
3	102	DBMS	B